

Skills Summary

Evaluating and Graphing Exponential Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluating a power rule

Order of work: in $a \cdot b^x$ the exponent belongs to b only. Compute b^x first, then multiply by a .

Two facts that carry most of the marks: $b^0 = 1$ for every nonzero base, so the graph always starts at a ; and $b^{-n} = \frac{1}{b^n}$, so a negative exponent gives a small *positive* value.

Example: For $f(x) = 4 \cdot 3^x$, find $f(2)$, $f(0)$ and $f(-1)$.

Step 1. The exponent sits on the 3, so each answer is a power of three multiplied by four — never four raised to anything.

Step 2. $3^2 = 9$, so $4 \cdot 9$; and $3^0 = 1$, so $4 \cdot 1$.

Step 3. $3^{-1} = \frac{1}{3}$, so $4 \cdot \frac{1}{3}$.

$$f(2) = 36, \quad f(0) = 4, \quad f(-1) = \frac{4}{3}$$

Try it: For $g(x) = 6 \cdot 2^x$, find $g(3)$ and $g(-1)$.

check: $g(3) = 48, g(-1) = \frac{3}{2}$

Skill 2 — Table to graph

Method: choose inputs on both sides of 0, evaluate, plot, then join with a *smooth curve* — exponential graphs are never made of straight segments.

Shape facts: the curve crosses the vertical axis at a ; if $b > 1$ it climbs to the right, and if $0 < b < 1$ it falls. It approaches the horizontal axis but never reaches it, because a power of a positive base is never zero.

Example: Build a table for $y = 3^x$ at $x = -1, 0, 1, 2$.

Step 1. Inputs either side of zero show both the flattening tail and the steep climb, which is what makes the curve recognisable.

Step 2. $3^{-1} = \frac{1}{3}$ and $3^0 = 1$.

Step 3. $3^1 = 3$ and $3^2 = 9$.

$\frac{1}{3}, \quad 1, \quad 3, \quad 9$ — plot and join smoothly.

Try it: For $y = 5 \cdot 2^x$, find the outputs at $x = -1, 0$ and 2 .

check: $2.5, 5, 20$

(!) Watch out: $4 \cdot 2^x$ is not 8^x , and 3^{-2} is not -9 . The coefficient stays outside the power, and a negative exponent flips the power over rather than changing its sign.

Skill 3 — Solving by matching powers

Method: to solve $a \cdot b^x = c$, divide both sides by a *first*, then rewrite the right-hand side as a power of b . Equal powers of the same base have equal exponents.

Powers worth knowing: powers of two run 2, 4, 8, 16, 32, 64; powers of three run 3, 9, 27, 81; powers of five run 5, 25, 125, 625.

Example: Solve $3 \cdot 4^x = 192$.

Step 1. The coefficient blocks the power comparison, so clear it before rewriting anything as a power.

Step 2. Divide by 3: $4^x = 64$.

Step 3. $64 = 4^3$, so the exponents match. Check: $3 \cdot 64 = 192$.

$$x = 3$$

Try it: Solve $2 \cdot 5^x = 1250$.

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